

heavy-ball-lean: Formal Proofs for Heavy Ball Linear Convergence in Lean 4

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Abstract

`heavy-ball-lean` is a Lean 4 / Mathlib library formalising a certified convergence argument for the heavy ball method, also known as Polyak momentum. The library works over a real inner product space, packages smoothness and strong-convexity assumptions, defines the classical constant parameters, records a momentum update and Lyapunov energy, and proves geometric decay of the certified energy and function-value gap. The method is a Lyapunov-energy proof of linear convergence for a certified heavy-ball orbit. The development is machine-checked in Lean 4 with zero `sorry`, zero `admit`, and standard Lean/Mathlib axioms only.

1. Introduction

The heavy ball method augments gradient descent with momentum. Its update uses both the current gradient and the previous displacement:

$$\begin{aligned} \mathbf{x}' &= \mathbf{x} - \alpha * \text{grad } f(\mathbf{x}) + \beta * \mathbf{p}, \\ \mathbf{p}' &= \mathbf{x}' - \mathbf{x}. \end{aligned}$$

For smooth strongly convex objectives, the classical parameter choice depends on the condition number $\kappa = L / \mu$ and gives a linear rate controlled by $\rho = (\sqrt{\kappa} - 1) / (\sqrt{\kappa} + 1)$.

The Lean library formalises the parameter algebra and the convergence consequence of a certified Lyapunov descent property. It does not derive the descent certificate from all analytic assumptions; instead, the orbit structure carries the update rule and the energy decrease property needed for the final convergence theorem.

2. Mathematical Setting

`HeavyBall/Defs.lean` defines `HeavyBallSetup` over a real inner product space E . The setup includes an objective f , gradient oracle `grad_f`, constants L and μ , a minimiser `xStar`, positivity and ordering hypotheses for L and μ , and smoothness and strong-convexity assumptions.

The classical parameters are defined as

```

alpha = 4 / (sqrt L + sqrt mu)^2
beta  = ((sqrt L - sqrt mu) / (sqrt L + sqrt mu))^2
kappa = L / mu
rho    = (sqrt kappa - 1) / (sqrt kappa + 1).

```

The Lyapunov energy combines objective suboptimality with a momentum-weighted squared norm:

```

E_k = f(x_k) - f(xStar)
      + mu / 2 * ||x_k + (1 / sqrt mu) * p_k - xStar||^2.

```

3. Main Theorems

The setup proves parameter facts including `sqrt_L_pos`, `sqrt_mu_pos`, `sqrt_mu_le_sqrt_L`, `one_le_kappa`, `rho_nonneg`, `rho_lt_one`, `rho_alt`, and `beta_eq_rho_sq`.

`Descent.lean` proves the certified descent theorem

```

heavy_ball_descent:
  E_{k+1} <= rho * E_k,

```

as well as `energy_nonneg` and `energy_geometric`:

```

E_k <= E_0 * rho^k.

```

`Convergence.lean` proves `fval_le_energy`, showing the function-value gap is bounded by the energy, and the main theorem `heavy_ball_convergence`:

```

f(x_k) - f(xStar) <= E_0 * rho^k.

```

4. Proof Sketch

The proof is organised around the Lyapunov energy. The parameter lemmas show that the rate is well formed: `rho` is nonnegative and strictly less than one, and `beta` equals `rho^2`. The `HeavyBall0orbit` structure supplies both the state recurrence and the descent certificate.

Given $E_{k+1} \leq \rho * E_k$, induction yields the geometric estimate $E_k \leq E_0 * \rho^k$. Nonnegativity of the squared norm term implies the function-value gap is bounded above by the energy. Combining those two results gives the final linear convergence theorem.

5. Relation to Sibling Libraries

`heavy-ball-lean` is closest to `nesterov-lean`, DOI 10.5281/zenodo.20474481, and `gradient-descent-lean`, DOI 10.5281/zenodo.20472996. Those libraries formalise first-order convergence without Polyak momentum. `contraction-lean`, DOI 10.5281/zenodo.20474762, supplies a related geometric-decay proof pattern, while `admm-lean` uses a Lyapunov decrease argument for residual convergence rather than momentum acceleration.

6. Conclusion

`heavy-ball-lean` formalises the parameter algebra and Lyapunov convergence shell for the heavy ball method. It proves that any certified orbit satisfying the stated energy descent condition has a geometrically decaying objective gap. Future work could derive the descent certificate directly from smoothness and strong convexity, compare the rate to Nesterov acceleration, and instantiate the framework for quadratic objectives.

References

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